

SolarGrid Demo Utility — Synthetic Grid Operating Code

SG-OP-001

SOLARGRID AI — Synthetic Engineering Demonstration Corpus

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Document Title	SolarGrid Demo Utility — Synthetic Grid Operating Code
Version	1.0
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Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY
Purpose	Define the synthetic operating envelope used by planning, validation, and safety demonstrations.

Demo Note: This document contains synthetic engineering requirements created for the SOLARGRID AI demonstration. All numerical values, thresholds, equipment ratings, and procedures are illustrative and must be replaced or validated by the deploying organization before operational use.

Use restriction: This document is intended only for RAG testing, semantic retrieval testing, SafetyAgent demonstration, engineering evidence demonstration, PlannerAgent integration testing, and Tool 16 testing. It is not intended for real-world grid operation.

Document Control

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Scope: Define the synthetic operating envelope used by planning, validation, and safety demonstrations.

Revision History

Revision	Date	Description	Author
1.0	2026-01-15	Initial synthetic demonstration release	SolarGrid Demo Utility — Engineering Department

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1. Purpose and Scope

This synthetic operating code defines the demonstration operating envelope for the SolarGrid Demo Utility electrical system. It establishes example operating states, reserve expectations, voltage and thermal limits, generator limits, BESS operating rules, and human approval points used by the SOLARGRID AI demonstration.

All limits in this document are configurable demonstration parameters. They are not legal, regulatory, grid-code, interconnection, or reliability requirements.

GRID-001 — The demonstration operating plan shall maintain sufficient generation and battery capability to meet the synthetic operating reserve requirement of 25 MW.

GRID-002 — Normal system operation shall maintain monitored bus voltages between 0.95 pu and 1.05 pu.

2. System Operating States

State	Entry Condition	Operating Objective	Planner/Safety Treatment
Normal	All monitored constraints within normal range	Meet load, reserve, voltage and thermal limits	Plan may proceed after required approval
Alert	A planning alert threshold is crossed without emergency condition	Restore emergency condition	Impact assessment required
Emergency	A synthetic emergency criterion is present	Protect equipment and maintain safe operating envelope	Safety review and corrective action required
Restoration	System returning from emergency	Re-establish normal margins	Human-supervised transition

GRID-003 — A transition from Normal to Alert shall trigger a documented review of the affected constraint and its margin.

GRID-004 — Emergency operating actions shall not be inferred from optimization feasibility alone; the applicable engineering evidence and human approval status shall be checked.

3. Normal Operating Conditions

Normal operation is the reference state for planning and validation. Forecasts, generator commitments, BESS SOC, network flows, and reserve are evaluated against the same synthetic operating envelope.

GRID-005 — The system shall maintain the minimum operating reserve of 25 MW under the validated Normal operating plan.

GRID-006 — No continuously monitored transmission or transformer element shall exceed 100% of its synthetic continuous rating in Normal operation.

4. Emergency Operating Conditions

Emergency conditions are demonstration states used to test conservative handling. Examples include a monitored line or transformer above its continuous limit, a voltage outside the Alert range, a reserve shortfall, or loss of a planned resource.

GRID-007 — When a required engineering constraint cannot be verified, the plan shall not be treated as validated solely because an optimization result is mathematically feasible.

Corrective actions may include redispatch, BESS response, commitment of another generator, forecast-driven replanning, or controlled load management in a separate approved procedure.

5. Frequency Operating Reserve

5.1 Reserve Definitions

Operating reserve is the amount of additional deliverable active power available within the defined response period. Generator headroom and BESS discharge capability may both contribute when they meet their respective availability, ramp, SOC, and response conditions.

RES-001 — The minimum total operating reserve requirement for the demonstration system is 25 MW.

RES-002 — Only reserve that is both technically deliverable and supported by the current equipment state may be counted toward the reserve requirement.

RES-003 — BESS reserve contribution shall respect the configured minimum SOC, reserve SOC, discharge power limit, and current SOC.

5.2 Reserve Example

Example: 18 MW of generator reserve plus 12 MW of eligible BESS reserve produces 30 MW total available reserve. The 25 MW requirement is therefore satisfied for this synthetic example, subject to all other constraints.

6. Voltage Limits

Parameter	Synthetic Limit	Meaning
Normal voltage	0.95–1.05 pu	Preferred operating range
Alert voltage	0.93–1.07 pu	Planning/operating review threshold
Emergency outside range	<0.93 pu or >1.07 pu	Corrective action and safety review

NET-001 — Normal operating bus voltage shall remain within 0.95–1.05 pu.

NET-002 — A bus voltage below 0.93 pu or above 1.07 pu shall be treated as an emergency-range violation for the demonstration.

7. Thermal Loading Limits

Transmission lines and transformers are monitored against synthetic continuous ratings. A planning alert threshold of 90% is used to provide margin before the 100% continuous limit.

NET-003 — The maximum continuous loading of a monitored line or transformer is 100% of its synthetic continuous rating.

NET-004 — Loading above 90% shall generate a planning alert requiring margin review.

8. Generator Operating Limits

Generator	Pmin (MW)	Pmax (MW)	Ramp Up (MW/min)	Ramp Down (MW/min)
G1	20	100	15	20
G2	15	80	12	16
G3	10	60	10	12

GEN-001 — Each committed generator shall operate no lower than its stated Pmin and no higher than its stated Pmax.

GEN-002 — Generator dispatch changes shall respect the applicable ramp-up and ramp-down limits.

9. Battery Operating Requirements

BESS-001 — During normal grid operation, the BESS SOC shall remain between 20% and 90%.

BESS-002 — The maximum continuous BESS discharge power shall not exceed 50 MW.

BESS-003 — The maximum continuous BESS charging power shall not exceed 40 MW.

BESS-004 — At least 15% of usable BESS energy capacity shall be reserved for emergency response unless an approved operating exception is active.

With a 120 MWh usable energy capacity, the 15% emergency reserve SOC corresponds to 18 MWh of stored energy. The 20% normal minimum SOC corresponds to 24 MWh.

10. Operator Approval Requirements

APP-001 — A plan affecting material generator commitment, BESS reserve, or network constraints shall have a human approval status before execution.

APP-002 — A plan with unresolved evidence insufficiency shall remain pending review and shall not be represented as engineering-approved.

11. Planning and Validation Requirements

VAL-001 — A planning result shall be validated against generator, BESS, reserve, voltage, thermal, forecast, and change-impact evidence applicable to the proposed operating state.

VAL-002 — If a required engineering evidence item is missing or unverifiable, the validation status shall be UNKNOWN rather than PASS.

This operating code is intentionally written so individual requirements can be replaced by customer-approved engineering documents without changing the SOLARGRID AI evidence architecture.